Sai Chaitanya Gelivi

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Education

University of Oklahoma — Norman, OK
Master of Science in Computer Science GPA: 3.91/4.00 — *Aug 2024 – May 2026*

Vardhaman College of Engineering — Hyderabad, India
Bachelor of Technology in Computer Science (AI & ML) GPA: 9.66/10.00 — *Dec 2020 – May 2024*

Professional Experience

MassMutual — Remote
Machine Learning Intern — *Aug 2022 – Jan 2023*

- Built and deployed ARIMA/SARIMA and LSTM time-series models on 3+ years of transactional data (100k+ records across timestamps, product categories, and geography), improving monthly ticket volume forecast accuracy by 20%.
- Engineered ETL pipelines for multi-dimensional data ingestion, cleaning, and feature extraction, cutting preprocessing time by 30% and enabling faster model iteration cycles.
- Delivered interactive Matplotlib dashboards and executive reports translating model outputs into actionable operational decisions for resource planning teams.

Technical Skills

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL, HTML5, CSS3
Frameworks & Libraries: Spring Boot, React, Angular, Node.js, Express.js, TensorFlow, Keras, Scikit-learn, Pandas, NumPy
AI/ML: Machine Learning, Deep Learning, Embeddings, Vector Search, Anomaly Detection, LLM/GenAI APIs
Databases: PostgreSQL, MySQL, MongoDB, Oracle (Relational & NoSQL)
Cloud: Amazon AWS, Google Cloud
Tools & Platforms: Git/GitHub, Docker, Linux, CI/CD (GitHub Actions), Postman, Jupyter, VS Code, IntelliJ
Concepts: Data Structures & Algorithms, OOP, REST APIs, Microservices, Secure Coding, Automated Testing, Agile/DevSecOps

Projects

Postgres Native Vector Retrieval | *C, Python, PostgreSQL, pgvector, HNSW, PL/pgSQL, Docker* — GitHub

- Authored a native PostgreSQL extension in C implementing HNSW-based approximate nearest neighbor search directly inside the database engine, eliminating the need for an external vector store such as Pinecone or Weaviate and reducing infrastructure complexity by removing a network hop from every query.
- Achieved sub-10ms semantic query latency on 100k+ embedding datasets with 768-dimensional vectors, delivering a 5–10x speedup over a baseline brute-force kNN scan while maintaining recall@10 above 0.95.
- Benchmarked retrieval performance across multiple HNSW index configurations (graph connectivity and search depth parameters) and containerized the full stack with Docker Compose, enabling one-command reproducibility and cutting evaluation setup time from hours to minutes.

TrackR — Full-Stack Job Application Tracker | *React, Node.js, MongoDB, JWT, Google OAuth 2.0* — GitHub | Live

- Built and deployed a full-stack MERN application tracking 10+ fields per job application with real-time search, filtering, CSV export, and dark mode; shipped to production on Vercel and Render with MongoDB Atlas as the database, serving a live public URL.
- Implemented hybrid authentication supporting both email/password (bcrypt-hashed credentials with JWT-based sessions) and Google OAuth 2.0, with automatic account linking via shared email matching so users can seamlessly switch between sign-in methods without losing their data.
- Designed the analytics dashboard using MongoDB aggregation pipelines ($group, $match, $cond) to compute response rates and status breakdowns server-side rather than pulling documents to the client, reducing payload size and keeping API responses consistent as the dataset scales.

TutorHub — Multi-Tenant SaaS Platform API | *Java 21, Spring Boot, Spring Data JPA, JUnit, Docker* — GitHub | Live

- Architected a multi-tenant REST API in Java/Spring Boot serving isolated tutoring academies, with JWT stateless authentication and a five-role RBAC model enforcing cross-tenant data isolation and per-user row-level access scoping.
- Designed a normalized PostgreSQL schema of 8+ entities via Spring Data JPA/Hibernate with Flyway migrations and optimistic locking for safe concurrent grading; added a scheduled async email-reminder service (Spring Scheduling + JavaMailSender) for assignments due within 24 hours.
- Validated with JUnit/Mockito and Testcontainers integration tests against real PostgreSQL (tenant-isolation and optimistic-lock cases), documented via OpenAPI/Swagger, and automated build/test/deploy with GitHub Actions and Docker to a live Render deployment.

Leadership & Honors

- **Vice President**, Competitive Coding Club, Vardhaman College of Engineering (2022–2024) — grew participation by 40%; recipient, Best Club Activities Award, IEEE Envision 2024.
- **Cultural Committee Member**, Indian Student Association (ISA), University of Oklahoma (2024–Present).